



Robotics are used to perform different types of automated tasks and for better production with good quality benefits

the growing demand for fresh and frozen goods, highlighted by Lineage Logistics' temperature-controlled warehouses. Omnichannel fulfilment strategies are being adopted to seamlessly integrate online and offline sales, a strategy employed by Target. Finally, flexible warehousing solutions, such as on-demand warehousing, are becoming popular, offering scalability and cost-efficiency, as evidenced by companies like Flexe. These trends are reshaping the warehousing industry, driving efficiency, sustainability, and responsiveness to market demands.

2. Cobots: Transforming traditional methods. Cobots, or collaborative robots, are revolutionising warehouse operations by working alongside human staff to enhance workflows without replacing them. Although they require a significant initial investment and infrastructure upgrades, they offer optimised processes within existing setups. Traditional warehouse methods, predominantly reliant on manual labour, are now being transformed by startups through advanced robotics. The adoption of automated guided vehicles (AGVs) and autonomous mobile robots (AMRs) is reducing manual labour reliance,

improving accuracy, and increasing safety for both staff and inventory. These technologies, including drones, goods-to-person robots, and driverless forklifts, automate crucial inventory management tasks such as picking, palletising, and transportation, marking a significant trend in warehouse automation.

US-based startup Ware is innovating with autonomous drones that navigate using indoor geolocation and AI, capturing inventory data with a 2D scanner, which is then securely stored and analysed in the cloud. This technology not only speeds up inventory processes but also digitises the warehouse environment. Similarly, Peer Robotics is developing mobile robots that interpret human gestures and commands, adept at navigating crowded spaces and enhancing inventory tracking. This automation across various tasks ensures rapid order handling and substantial ROI gains, underscoring the critical role of robotics in the digital transformation of warehouses.

3. AI-driven WMS solutions. Warehouse management systems (WMS) enhance operations by offering real-time oversight from goods' arrival to departure, employing big

data and AI for detailed analytics. These systems improve visibility, accuracy, and speed while reducing costs and enhancing flexibility, safety, supplier relationships, and customer service.

Indian startup Tradexa has launched Hyper Inventory, an advanced multi-location warehouse management software that centralises control over pricing, inventory, and deliveries. It utilises AI-driven business intelligence for forecasting and performance analytics. Similarly, Chinese startup Multiway's WMS enhances control and visibility with features like real-time status updates and stock management, thereby reducing errors and increasing efficiency.

Walmart integrates real-time sales data with demand forecasting in its inventory management system, optimising stock levels and enhancing ROI and customer satisfaction by ensuring product availability.

To combat inventory losses from spoilage, damage, or theft, warehouse managers rely on inventory tracking solutions. Technologies like radio frequency identification (RFID), which can read tags remotely and simultaneously, provide significant advantages over traditional barcoding systems by enhancing the accuracy and transparency of inventory management. Additionally, optical character recognition (OCR) technology speeds up warehouse processes by converting documents into digital formats for easy database integration.

Indian startup Octorite Technologies introduces an RFID scanning system equipped with tunnel technology and exclusive antennas, capable of reading 1000 tags per second, which greatly streamlines scanning processes for high-volume environments. In Nigeria, startup Fiber offers a free inventory management app that facilitates inventory tracking, sales recording, and receipt